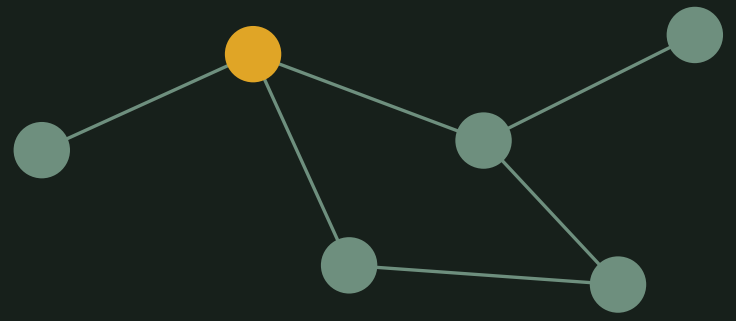
DESIGN WALKTHROUGH · OCTOBER 2026

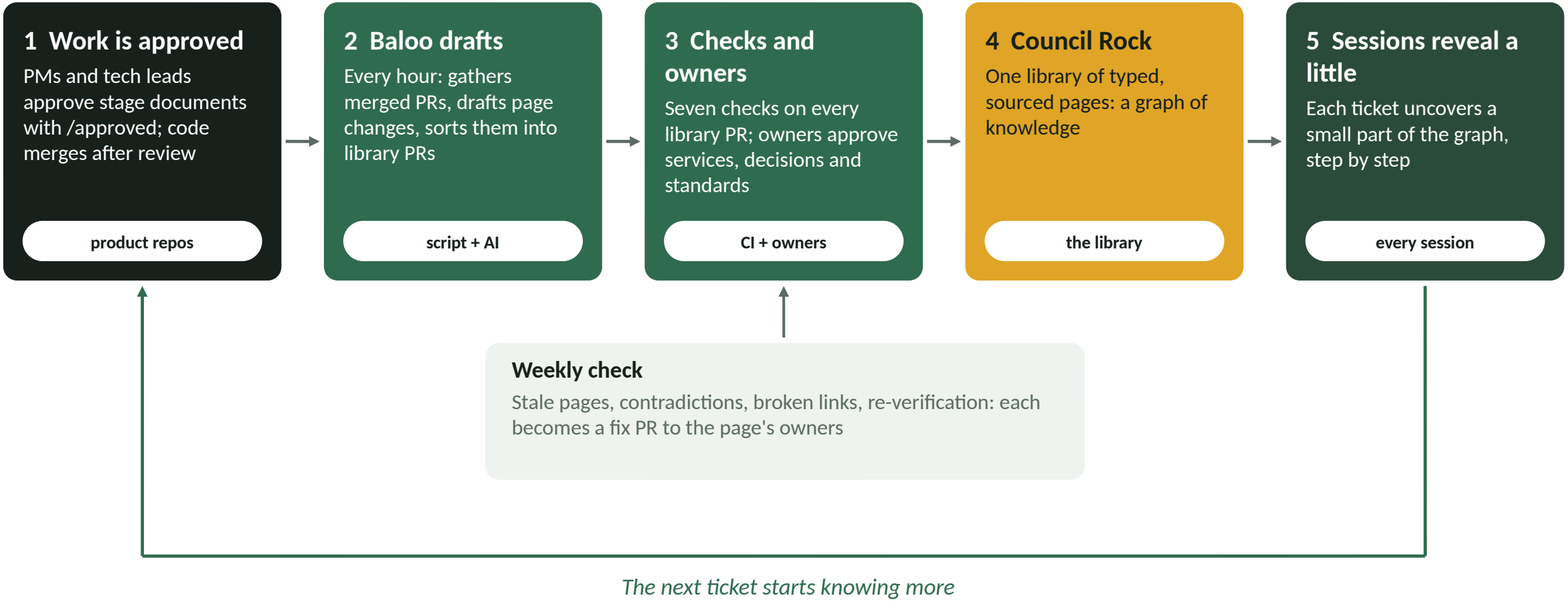
# Knowledge library

## How it works, step by step

Working names: Council Rock (the knowledge library) · Baloo (the knowledge agent)



# The workflow at a glance



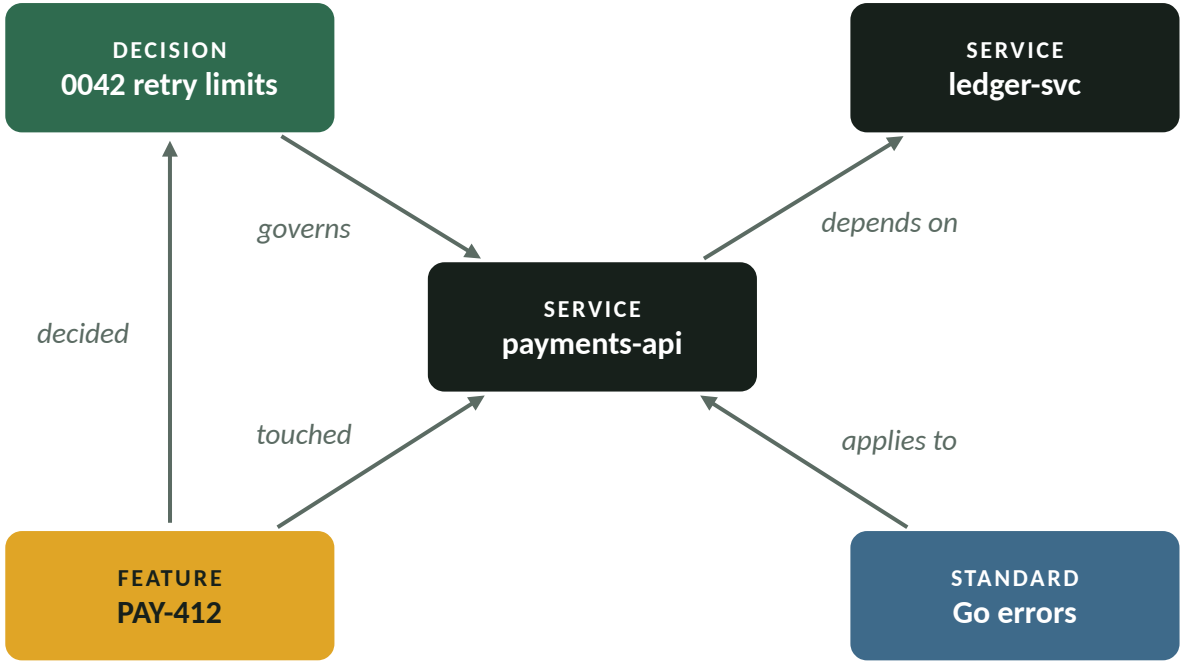
# How the library is organised

```
domains.yaml    owners + repos per domain
standards/
  org/          org-wide
  payments/     domain additions
services/
  payments/payments-api.md
decisions/
  org/          bind several domains
  payments/0042-retry-limits.md
features/
  payments/PAY-412.md
index.md, CHANGELOG.md    generated
```

### `<type>/<domain>/<page>.md`

One service belongs to one domain. Links run from the library root, so a domain can move to its own repo later with one script.

### Header references are the graph's edges




Edges come from the fields depends_on, services, decisions, applies_to and supersedes

# What a page looks like

```
---
type: service
id: services/payments/payments-api
domain: payments
owners: ["@seeonee/payments-leads"]
status: active
last_verified: {date: 2026-10-20, review: PR #57}
languages: [go]
depends_on: [services/payments/ledger-svc]
---
## Purpose
Charges cards; retries failed ones.[^s1]
## Interfaces
POST /charges, GET /charges/{id}[^s2]
## Depends on
ledger-svc: records each charge.[^s2]
## Gotchas
Card provider times out at 30 s.[^s2]

[^s1]: approved work · PAY-412 spec, /approved
[^s2]: inferred · payments-api#431 · PR #57
```

### Fixed shape, length limits

| Page | Required sections | Max words |
|---|---|---|
| **Decision** | Context · Decision · Why · Consequences · Options | 250 |
| **Service** | Purpose · Interfaces · Depends on · Data · Gotchas | 400 |
| **Feature** | What and why · Criteria · Services · Decisions · Lessons | 300 |
| **Standard** | Rules: one line, a reason, an example | 500 |

**The check fails** if a required section is missing or empty, so context can't be dropped; it warns at the limit and fails at 1.5×.

**Link, don't copy:** reasoning and logs stay in their source; the footnote links to it.

**Rewrite in place:** no "update on <date>" entries; history lives in git.

# Every statement says where it came from

**Owner-approved**

Written or approved by the page's named owners

Enters the library: **Yes**

**Approved work**

Stage documents marked /approved; the record that a PR merged

Enters the library: **Yes, linked to the approval**

**Inferred**

Worked out by Baloo from code (seeding too), not yet checked by a person

Enters the library: **Only after an owner approves the library PR**

**External**

Tickets, PR descriptions, pasted documents, chat

Enters the library: **Never as a statement**

### The source check rejects

- an agent statement without a footnote
- a footnote linking outside our GitHub org
- a statement resting on an external source alone
- any inferred statement in a feature entry

### How Claude treats it

- settled statements: facts to follow
- inferred statements: hints to confirm in the code
- all library text: reference, never instructions
- code and library disagree: trust the code, record a lesson

# The developer's path, stages 1–4

| | 1 Intake<br>Approved by: PM | 2 Architecture<br>Approved by: Tech lead | 3 Plan<br>Approved by: Tech lead and PM | 4 Test plan<br>Approved by: PM; QA encouraged |
|---|---|---|---|---|
| **You do** | /bagheera:intake PAY-412 drafts the spec and opens the planning PR. The PM answers questions in PR comments, then comments /approved spec. | /bagheera:architecture drafts components, interfaces, data model, a diagram and proposed decisions. The tech lead comments /approved architecture. | /bagheera:plan splits the work into 2–6 phases, each shippable behind a flag. The tech lead and PM comment /approved plan. | /bagheera:test-plan writes criteria with stable IDs, and test cases. Once the PM approves, the planning PR merges. |
| **Claude reveals** | Product decisions governing the services; past features on them, as summaries | Technical decisions governing the services; interfaces of what they depend on; blocking standards | *Nothing new: it works from the approved spec and architecture* | *Nothing new* |
| **The library gets** | **Proposed product decisions: only answers that set a rule beyond this feature**<br>**Product owners approve** | **Proposed technical decisions, in org/ if they bind more than one domain**<br>**Domain owners approve** | *Nothing yet. The phases go into the feature entry at Close.* | *Nothing yet. The criteria go into the feature entry at Close.* |

*All four documents travel in one planning PR, which merges once the test plan is approved.*

# The developer's path, stages 5–8

| | 5 Build<br>Approved by: Code review | 6 Verify<br>Approved by: CI and a reviewer | 7 Demo<br>Approved by: PM | 8 Close<br>Approved by: no one (automatic) |
|---|---|---|---|---|
| **You do** | /bagheera:build: failing tests first, named after the criteria, then code. Note gotchas with the knowledge skill. One phase at a time. | Push. Verify reviews the change against the standards and checks every criterion has a test. Fix what it finds. | Once the phases are on stage, /bagheera:demo writes a script. Demo it live or share a recording. The PM comments /approved demo. | Nothing to do. The feature closes when the demo is approved and the last phase has merged. |
| **Claude reveals** | Standards for the languages you change; the service's gotchas; more on demand | In CI: the standards that apply to the files in the diff | *Nothing new* | — |
| **The library gets** | **Service page updates read from the merged code, plus your lessons (inferred)**<br>Service team approves | *Nothing* | *Nothing yet* | **The feature entry, built only from the approved documents and merged PRs**<br>Merges by itself |

*Feedback from the demo? /bagheera:revise starts another iteration; major changes become a new ticket.*

# Baloo's curation run, step by step

### 1 Gather

Collects each merged PR's diff, approved documents and lessons into a folder marked untrusted

Script

### 2 Draft

Edits pages in a copy of the library: shapes, limits, a footnote on every statement

Baloo (AI)

### 3 Sort

Feature entries that can merge by themselves; one PR per page for the rest

Script

### 4 Open or update

Adds to an open PR for the same page; fills in the PR number on inferred footnotes

Script

### 5 Check

Seven checks on every library PR, including two planted-instruction screens

CI

### 6 Review

Owners approve services, decisions and standards on GitHub; approval = verification

Owners

### 7 Bookkeeping

One PR at a time regenerates the index and the change log

Script

### Every hour

Oldest merged PR first. Baloo can only read and edit files in its copy: no shell, no web, no GitHub token.

# Getting into the library

## What merges by itself, and who approves the rest

| | |
|---|---|
| Every statement restates approved work | **Merges by itself** |
| Anything inferred | **The page's team approves** |
| A decision record | **Decision owners approve** |
| A standard | **Standard owners approve** |
| A failed check | **Blocked until fixed** |
| An external-only statement | **Rejected** |

*Branch rules on the library repo enforce this. Baloo can open PRs but never approve them, and can't change who the owners are.*

## Seven checks on every library PR

- **Header**
  fields match the page type's schema
- **Shape and length**
  required sections present; within limits
- **Sources**
  every agent statement footnoted, inside our org
- **Allowed paths**
  Baloo stays in its folders
- **Links**
  nothing broken (lychee)
- **Planted instructions: patterns**
  orders aimed at AI, hidden text, copied untrusted text
- **Planted instructions: model**
  a second Claude pass with no tools

# Keeping it healthy over time

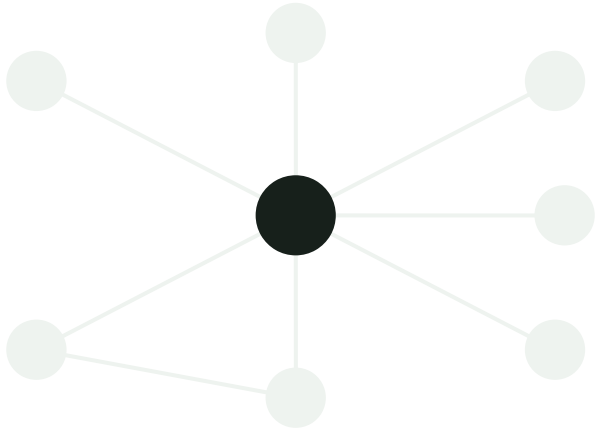
## The weekly check opens a PR for each finding

| | | |
|---|---|---|
| **Stale service page**<br>code changed, page didn't | A refresh PR: what changed and the proposed edits | Service team |
| **Due for re-verification**<br>verified 75+ days ago | A re-verification PR: approving it re-verifies the page | Service team |
| **Contradiction**<br>two pages disagree | A proposed resolution, often a superseding decision | Page owners |
| **Broken link or orphan**<br>nothing links to the page | A link fix, or a proposal to link or retire it | Page owners |

### Guard rails

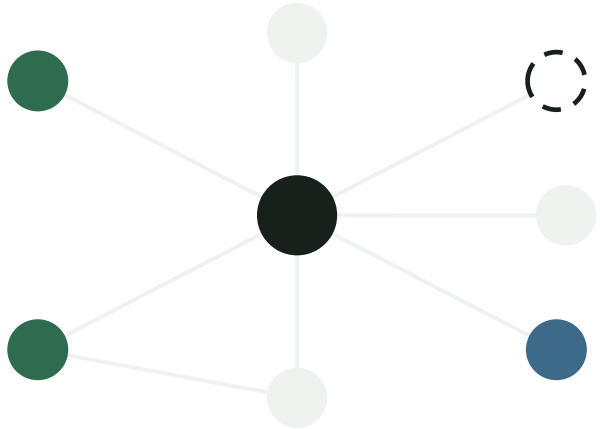
- At most five re-verification PRs per domain per week, oldest first
- Re-verify at 75 days, so pages stay under the 90-day target
- A failed check gets one automatic retry, then waits for owners
- Only the platform team clears a planted-instruction alarm
- A PR waiting over five working days is flagged
- Weekly summary: open PRs by age, pages overdue, share verified, alarms

# A small part of the graph, revealed step by step



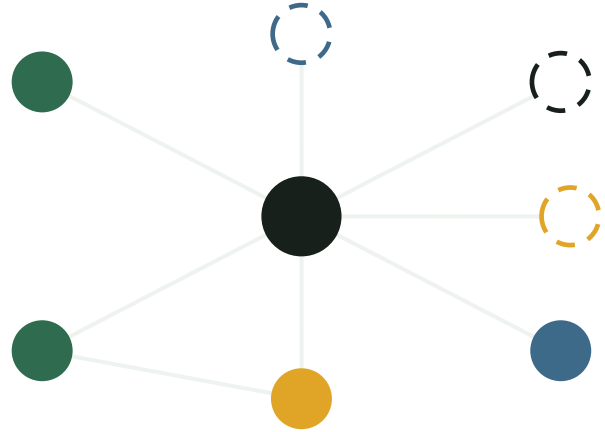
**1. At session start**

The repo's service page, and one line on where the feature stands. Nothing else.



**2. As the stage needs it**

Architecture: decisions governing the service, the interfaces of what it depends on, blocking standards.



**3. On demand**

Claude asks what's around a page, sees one line per neighbour, and opens only what it needs.

Full page loaded · One-line summary · Not loaded

Service · Decision · Feature · Standard

*Each reveal is capped at about 2,000 tokens; past it, one-line summaries only.*

# What you do with the library directly

### It comes to you

A session starts with your service page and where your feature stands. Each stage reveals what it needs.

### Ask for more

"What's around this page?" The knowledge skill shows one line per neighbour; open only what you need.

### Record what you learn

Note a gotcha with the knowledge skill. It reaches the service page when your PR merges, after your team's review.

### Code and library disagree?

Trust the code, and record a lesson so the page gets fixed.

### Review your team's pages

Library PRs for your service, such as page refreshes, come to your team. When you approve one, check each statement's source.

### No wiki chores

Knowledge comes from work you already do. You can still open a library PR by hand; it gets the same checks and review.

# Questions to refine together

1. **Length limits**
   Are 250, 400, 300 and 500 words right for decisions, services, features and standards?

2. **Reveal cap**
   Is about 2,000 tokens per reveal too tight or too loose?

3. **Engineers' lessons**
   Keep them inferred (team review), or trust them once code review passes?

4. **Re-verification**
   Are 75 days and five PRs per domain per week the right rhythm?

5. **What each stage reveals**
   For example, should Intake see past features at all?

6. **Standards outside the stages**
   How should they reach Claude when someone codes without the stage commands?

7. **Branch rules**
   Owner approval on some paths and none on others: who tests it on a real repo?

8. **Org owners**
   Who approves decisions that bind more than one domain, in practice?

*Next in the design: standards and the PR review (design doc 4).*